

Dexter Fernandes

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Dear Hiring Team,

I'm applying for the Software Engineer role at One Big Circle.

AIVR has to keep running unattended, even when nobody can reach in a hurry. That's the problem I spent three years on. At Esbaar I owned an NVIDIA DeepStream inference pipeline running around the clock on BP rigs in Oman, hours from the nearest engineer, where a silent failure meant lost data nobody caught until it hurt.

At Esbaar, the first version of our application leaked memory until it crashed, everytime. I wrote a watchdog to catch it and restart, but that was a plaster. The real fix was splitting the app into microservices and rewriting the streaming logic so it stopped eating RAM. I kept the watchdog anyway. When you can't reach the hardware, you want both.

Working the cloud end changed what I built on the device. At LivNSense, I designed the PostgreSQL schema behind the worker heat maps and deployed the Azure services feeding it. Helmet-colour classification stayed in the cloud because nothing waited on it. Tracking and fall detection went on the device, because sending events beats sending frames when the link is the constraint. I built that perception layer in C++17 with OpenVINO, across RGB and infrared streams.

I finished my MSc Robotics in Bristol in 2024, and I've kept building. Most recently a multi-stream DeepStream 9.0 pipeline with a custom C++/CUDA TensorRT plugin. Link's on my CV.

I would like to know where the edge/cloud boundary sits on the vehicle now. I'm in Bristol and free at short notice.

Kind regards,
Dexter Fernandes